

- RAG Learning

2.1 Algebraic Notation & Manipulation

Pearson Edexcel GCSE Mathematics 1MA1 — Algebra — spec refs A1–A4 — Papers 1 (non-calculator), 2 and 3 (calculator)

2.1.1 Algebraic Notation (A1)

In algebra a letter stands for a number that may vary or may be unknown. Edexcel mark schemes expect answers written with the standard shorthand below, and a final answer left as, for example, $3 \times a \times a$ instead of $3a^2$ can lose the accuracy mark.

Long form	Standard form	Comment
$p \times q$	pq	No multiplication sign; letters normally in alphabetical order
$m + m + m + m$ or $4 \times m$	$4m$	The number (coefficient) is written first
$t \times t$	t^2	" t squared"
$t \times t \times t$	t^3	" t cubed"
$p \times p \times q$	p^2q	The index belongs to p only
$m \div n$	$\frac{m}{n}$	Division is written as a fraction
$0.25y$	$\frac{1}{4}y$ or $\frac{y}{4}$	Coefficients are given as fractions rather than decimals

$5 \times (y - 1)$	$5(y - 1)$	A bracket shows the whole of $y - 1$ is multiplied by 5
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Exam Tip: Watch where an index sits. $3m^2$ means $3 \times m \times m$, while $(3m)^2 = 3m \times 3m = 9m^2$. Likewise $-x^2$ with $x = 4$ is -16 , but $(-x)^2$ is 16.

2.1.2 Substituting into Expressions and Formulae (A2)

To **substitute**, replace every letter by its value and evaluate using the order of operations (brackets, indices, then division and multiplication, then addition and subtraction). Edexcel often uses formulae from science, finance or everyday life. You do not need to know them in advance, because everything you need is given in the question.

Worked Example: Work out the value of $2a^2 - 5b$ when $a = -3$ and $b = 4$.

$$a^2 = (-3)^2 = 9 \text{ (square first, and the result is positive)}$$

$$2 \times 9 - 5 \times 4 = 18 - 20$$

$$= -2$$

Worked Example: The power, P watts, used by an electrical component is given by $P = I^2 R$. Work out P when $I = 3$ and $R = 12$.

$$I^2 = 3^2 = 9$$

$$P = 9 \times 12 = 108 \text{ watts}$$

Exam Tip: Write the substitution out in full, with negative numbers in brackets, before you calculate. On Paper 1 this working earns the M1 even if the arithmetic slips.

2.1.3 Expressions, Equations, Formulae, Identities and Inequalities (A3)

Term: one part of an expression, made from numbers and letters multiplied together, e.g. $-6y$, $4x^2$, 9 .

Coefficient: the number multiplying a letter, e.g. in $-7x^2$ the coefficient of x^2 is -7 .

Factor: a number, letter or bracket that divides exactly, e.g. 5 , y and $5y$ are factors of $10y$; $(x + 3)$ is a factor of $x^2 + 7x + 12$.

Expression: terms combined with no equals sign, e.g. $6y + 1$.

Equation: a statement that two expressions are equal, true only for certain values, e.g. $4y - 3 = 9$.

Formula: a rule connecting different variables, e.g. $F = ma$.

Identity: true for *all* values of the letters, shown with $\equiv$, e.g. $2(y - 3) \equiv 2y - 6$.

Inequality: a statement using $<$, $>$, $\leq$ or $\geq$, e.g. $y + 2 > 5$.

Worked Example: Name each statement: (A) $6y + 1$ (B) $4y - 3 = 9$ (C) $F = ma$ (D) $2(y - 3) \equiv 2y - 6$ (E) $y + 2 > 5$.

A: no equals sign → **expression**

B: only true when $y = 3$ → **equation**

C: links force, mass and acceleration → **formula**

D: true whatever y is → **identity**

E: uses $>$ → **inequality**

2.1.4 Collecting Like Terms and the Laws of Indices (A4)

Like terms

Terms are **like terms** when they contain exactly the same letters with the same powers, for example $5ab$ and $-2ba$. Add or subtract their coefficients; unlike terms such as $3m$ and $3m^2$ stay separate.

Worked Example: Simplify $8m + 3n - 5m - 7n + 2$.

$$m \text{ terms: } 8m - 5m = 3m$$

$$n \text{ terms: } 3n - 7n = -4n$$

$$\text{Answer: } 3m - 4n + 2$$

Laws of indices

$$x^a \times x^b = x^{a+b} \quad x^a \div x^b = x^{a-b} \quad (x^a)^b = x^{ab}$$

$$x^0 = 1 \quad x^{-a} = \frac{1}{x^a}$$

When terms have coefficients, deal with the numbers and each letter separately: multiply (or divide) the numbers, then add (or subtract) the powers letter by letter.

Worked Example: Simplify $(2x^3y)^4 \div 4x^5y^2$.

$$(2x^3y)^4 = 2^4 \times x^{12} \times y^4 = 16x^{12}y^4 \text{ (the 2 is raised to the power 4 too)}$$

$$16 \div 4 = 4, \quad x^{12-5} = x^7, \quad y^{4-2} = y^2$$

$$\text{Answer: } 4x^7y^2$$

Worked Example: Simplify $6x^{-2} \times 3x^7$.

$$6 \times 3 = 18 \text{ and } -2 + 7 = 5$$

$$\text{Answer: } 18x^5$$

2.1.5 Expanding a Single Bracket and Taking Out Common Factors (A4)

Expanding

The term outside multiplies *each* term inside: $k(a - b) = ka - kb$. A negative term outside changes the sign of everything inside.

Worked Example: Expand and simplify $5(2x + 3) - 3(x - 4)$.

$$5(2x + 3) = 10x + 15$$

$$-3(x - 4) = -3x + 12 \quad (-3 \times -4 = +12)$$

$$10x + 15 - 3x + 12 = 7x + 27$$

Factorising with a common factor

Factorising reverses expanding. Find the **highest common factor** of the numbers and of each letter, put it outside the bracket, and divide every term by it to fill the bracket.

Worked Example: Factorise fully $9a^2b + 15ab^3$.

HCF of 9 and 15 is 3; lowest power of a is a ; lowest power of b is b

$$\text{HCF} = 3ab$$

$$9a^2b \div 3ab = 3a \text{ and } 15ab^3 \div 3ab = 5b^2$$

Answer: $3ab(3a + 5b^2)$

Exam Tip: "Factorise fully" means nothing more can be taken out. An answer such as $3(3a^2b + 5ab^3)$ is correct but incomplete and typically scores M1 A0.

2.1.6 Expanding Products of Two Binomials (A4)

A **binomial** has two terms, for example $(2x - 5)$. Multiplying two binomials gives four products, which you then simplify by collecting like terms. A grid keeps the four products in order.

Grid method: $(3x + 2)(2x - 5)$

	2x	-5	
3x	6x ²	-15x	$\begin{array}{r} 6x^2 - 15x + 4x - 10 \\ \hline = 6x^2 - 11x - 10 \end{array}$
+2	+4x	-10	

Each cell = row term × column term

Worked Example: Expand and simplify $(x - 4)(x + 9)$.

$$x \times x = x^2, \quad x \times 9 = 9x, \quad -4 \times x = -4x, \quad -4 \times 9 = -36$$

$$x^2 + 9x - 4x - 36 = x^2 + 5x - 36$$

Squares and the difference of two squares

$$(a + b)^2 = a^2 + 2ab + b^2 \quad (a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2$$

Worked Example: Jo writes $(x - 3)^2 = x^2 - 9$. Show that Jo is wrong and give the correct expansion.

Try $x = 5$: $(5 - 3)^2 = 4$, but $5^2 - 9 = 16$. The two sides are not equal, so Jo is wrong.

$$(x - 3)(x - 3) = x^2 - 3x - 3x + 9 = x^2 - 6x + 9$$

Exam Tip: Always write a squared bracket out twice before expanding. Forgetting the middle term ($2ab$) is the commonest error on these questions.

2.1.7 Factorising $x^2 + bx + c$ and the Difference of Two Squares (A4)

For $x^2 + bx + c$, look for two numbers that **multiply to** c and **add to** b . They become the numbers in the two brackets.

Worked Example: Factorise $x^2 + 10x + 21$.

Factor pairs of 21: 1×21 , 3×7

$$3 + 7 = 10 \checkmark$$

Answer: $(x + 3)(x + 7)$

Worked Example: Factorise $x^2 - 4x - 21$.

Product -21 , sum -4 : the numbers are -7 and $+3$

Answer: $(x - 7)(x + 3)$ (check: $-7x + 3x = -4x \checkmark$)

Signs in $x^2 + bx + c$

Signs in the brackets

Example

b and c both positive	$(x+ \quad)(x+ \quad)$	$x^2 + 8x + 15 = (x + 3)(x + 5)$
b negative, c positive	$(x- \quad)(x- \quad)$	$x^2 - 8x + 15 = (x - 3)(x - 5)$
c negative	one $+$ and one $-$	$x^2 - 2x - 15 = (x - 5)(x + 3)$

Difference of two squares

Two square terms with a minus sign between them and no middle term factorise as

$$a^2 - b^2 = (a + b)(a - b)$$

Worked Example: Factorise $x^2 - 121$.

$$x^2 - 11^2 = (x + 11)(x - 11)$$

Worked Example: Factorise fully $2x^2 + 4x - 30$.

Take out the common factor 2: $2(x^2 + 2x - 15)$

Product -15 , sum $+2$: $+5$ and -3

Answer: $2(x + 5)(x - 3)$

Exam Tip: Look for a common factor before anything else, then expand your brackets to check. A quick check catches almost every sign error.

2.1.8 Expanding Brackets Involving Surds (A4)

The 1MA1 specification includes expressions involving surds in A4 at both tiers. Treat a surd like a letter when expanding, and use $\sqrt{a} \times \sqrt{a} = a$ to tidy up. (Simplifying surds such as $\sqrt{12} = 2\sqrt{3}$ and rationalising denominators are Higher-tier skills from N8.)

Worked Example: Expand and simplify $(\sqrt{7} + 3)(\sqrt{7} - 1)$.

$$\sqrt{7} \times \sqrt{7} = 7, \quad \sqrt{7} \times (-1) = -\sqrt{7}, \quad 3 \times \sqrt{7} = 3\sqrt{7}, \quad 3 \times (-1) = -3$$

$$7 - \sqrt{7} + 3\sqrt{7} - 3 = 4 + 2\sqrt{7}$$

Worked Example: Expand $(1 + \sqrt{3})^2$.

$$(1 + \sqrt{3})(1 + \sqrt{3}) = 1 + \sqrt{3} + \sqrt{3} + 3$$

$$= 4 + 2\sqrt{3}$$

Brackets in the form $(a + \sqrt{b})(a - \sqrt{b})$ give a whole number, because the surd terms cancel:
 $(\sqrt{6} + 2)(\sqrt{6} - 2) = 6 - 4 = 2$.

2.1.9 Factorising $ax^2 + bx + c$ and Expanding Three or More Binomials (A4) Higher

Factorising when the coefficient of x^2 is not 1

1. Multiply a by c .
2. Find two numbers with product ac and sum b .
3. Use them to split the bx term into two terms.
4. Factorise each pair, then take out the shared bracket.

Worked Example: Factorise $4x^2 + 4x - 15$.

$$ac = 4 \times (-15) = -60; \text{ product } -60 \text{ and sum } 4: +10 \text{ and } -6$$

$$4x^2 + 10x - 6x - 15$$

$$2x(2x + 5) - 3(2x + 5)$$

Answer: $(2x - 3)(2x + 5)$

The difference of two squares extends to terms with coefficients: $25x^2 - 9 = (5x + 3)(5x - 3)$.

Products of three binomials

Multiply two of the brackets together and simplify, then multiply every term of that result by each term of the third bracket.

Worked Example: Expand and simplify $(x + 1)(x - 3)(x + 5)$.

$$(x + 1)(x - 3) = x^2 - 2x - 3$$

$$(x^2 - 2x - 3)(x + 5) = x^3 + 5x^2 - 2x^2 - 10x - 3x - 15$$

$$= x^3 + 3x^2 - 13x - 15$$

Exam Tip: Check a cubic expansion by substituting $x = 1$ into both forms: $(2)(-2)(6) = -24$ and $1 + 3 - 13 - 15 = -24$ ✓.

2.1.10 Algebraic Fractions (A4) Higher

Simplifying

Factorise the top and bottom completely, then cancel common **factors** (whole brackets). You can never cancel separate terms that are added or subtracted.

Worked Example: Simplify fully $\frac{x^2 - 9}{2x^2 + 5x - 3}$.

$$\text{Numerator: } (x + 3)(x - 3)$$

$$\text{Denominator: } ac = -6, \text{ numbers 6 and } -1: 2x^2 + 6x - x - 3 = 2x(x + 3) - 1(x + 3) = (2x - 1)(x + 3)$$

Cancel $(x + 3)$: $\frac{x - 3}{2x - 1}$

Adding and subtracting

Write both fractions over a common denominator, multiply each numerator accordingly, then simplify the numerator.

Worked Example: Write $\frac{3}{x+2} - \frac{1}{x-4}$ as a single fraction in its simplest form.

Common denominator: $(x + 2)(x - 4)$

$$\frac{3(x - 4) - (x + 2)}{(x + 2)(x - 4)}$$

Numerator: $3x - 12 - x - 2 = 2x - 14 = 2(x - 7)$

Answer: $\frac{2(x - 7)}{(x + 2)(x - 4)}$

Multiplying and dividing

Factorise everything first. To divide, multiply by the reciprocal of the second fraction, then cancel.

Worked Example: Simplify $\frac{x^2 + 5x + 6}{x^2 - 4} \div \frac{x + 3}{3x - 6}$.

$$\frac{(x + 2)(x + 3)}{(x + 2)(x - 2)} \times \frac{3(x - 2)}{x + 3}$$

Cancel $(x + 2)$, $(x + 3)$ and $(x - 2)$

Answer: 3

Exam Tip: Put the second numerator in brackets when subtracting: $-(x + 2) = -x - 2$. Missing this sign change costs the accuracy mark more often than any other slip.

Quick Reference Summary

Skill	Key rule
Notation (A1)	$pq, 4m, p^2q, \frac{m}{n}$; fractional coefficients; simplest form
Substitution (A2)	Brackets round negatives; indices before multiplying
Vocabulary (A3)	Expression, equation, formula, identity ($\equiv$), inequality, term, factor
Indices (A4)	Add powers to multiply, subtract to divide, multiply for a power of a power
Single bracket / HCF (A4)	$k(a - b) = ka - kb$; take out the HCF of numbers and letters
Two binomials (A4)	Four products then collect; $(a \pm b)^2 = a^2 \pm 2ab + b^2$
$x^2 + bx + c$ (A4)	Two numbers that multiply to c and add to b
Difference of two squares (A4)	$a^2 - b^2 = (a + b)(a - b)$
Surds in brackets (A4)	Expand as normal; $\sqrt{a}\sqrt{a} = a$
$ax^2 + bx + c$ Higher	Split the middle term using ac , then factorise in pairs
Three binomials Higher	Expand two, simplify, then multiply by the third
Algebraic fractions Higher	Factorise fully and cancel brackets; common denominator to add or subtract